

Human–Computer Interaction (HCI)

Full-Length Practice Exam + Marking Scheme + Model Answers

Time: 2.5–3 hours | Total Marks: 100 | Date: 15 Dec 2025

Instructions

- Attempt the paper in order. Use correct HCI terminology (principle names and evaluation method names).
- For design questions: (1) name the principle/heuristic, (2) point to the exact UI element/problem, (3) explain impact, (4) propose a concrete fix.
- For evaluation questions: include user group, method, tasks, success metrics (effectiveness, efficiency, satisfaction), data (quant + qual), and limitations.
- Use keywords where relevant: direct manipulation, rapid and reversible actions, visibility of system status, recognition rather than recall, informative feedback, closure, internal locus of control, universal usability, mental model, cognitive load.

PART A — Short Answer Theory (30 marks)

Answer ALL. Each answer should be ~8–12 lines and must use correct terms.

Q1. (6 marks) Define usability testing and list the three goals of usability testing taught in class.

Q2. (6 marks) Differentiate between Shneiderman's 'offer informative feedback' and Nielsen's 'visibility of system status'. Give one UI example for each.

Q3. (6 marks) Explain 'support internal locus of control' and describe two UI features that strengthen it in a transactional app.

Q4. (6 marks) What does 'reduce short-term memory load' mean? Explain recognition vs recall and give two interface techniques.

Q5. (6 marks) Explain iterative design and why 'one study isn't truth' (triangulation). How does this change design decisions?

PART B — Design Analysis (Identify violations) (20 marks)

For EACH scenario: list at least 6 problems, map each to a named principle/heuristic, and explain impact on the user.

Q6. (10 marks) Scenario: **Food Delivery Checkout Screen**. Analyze the design.

- The screen shows 14 sections in one scroll: address, coupon, payment, tips, delivery time, cutlery option, donation, special instructions, rider note, gift wrap, loyalty points, etc.
- 'Place Order' button appears only at the very bottom after long scrolling.
- Coupon is a small text link ('Apply promo') hidden under 'More options'.
- Error message after payment failure says: 'Something went wrong (Code 14)'.
- Back button exits to home and clears cart without warning.
- Total price changes after you press 'Place Order' because 'service fee' appears later.

Q7. (10 marks) Scenario: **University Portal Login + OTP**. Analyze the design.

- Login asks for 'User ID' with no hint; students use ERP, staff use email, but not mentioned.
- Password rules are not shown; after error, it says 'invalid format'.
- OTP screen has a 30-second timer but no resend button; after expiry, the only option is 'Start over'.
- 'Remember me' checkbox is on OTP screen, not on login.
- Keyboard Enter submits early and clears OTP fields.
- Error message is red text at top; focus stays at bottom.

PART C — Fix-the-Design (Apply principles) (20 marks)

For EACH prompt: propose fixes. Mention target principle(s), exact UI changes, and why they improve usability.

Q8. (5 marks) Fix the design: **Ride-hailing 'Schedule Ride' flow.**

- Users can schedule a pickup, but the time picker defaults to 3:00 AM and shows a tiny AM/PM toggle.
- There is no summary screen; pressing 'Confirm' instantly books.
- Users often schedule the wrong day because the calendar is hidden behind a small icon.
- Cancellation is buried in Settings.

Q9. (5 marks) Fix the design: **E-commerce returns form.**

- Return reason is a long dropdown of 40 items with similar wording.
- Users must type order number from memory; the app shows no recent orders list.
- Photo upload is optional but error says 'Upload image to continue'.
- After submitting, there is no tracking number or confirmation message.

Q10. (5 marks) Fix the design: **Banking app money transfer.**

- Beneficiary list is sorted randomly. Search is only by account number.
- Transfer limit and fee are shown only after 'Next'.
- If transfer fails, user is returned to home screen and cannot see what went wrong.
- There is no 'Edit' step before final submission.

Q11. (5 marks) Fix the design: **Hospital appointment kiosk (touchscreen).**

- Font is small; important buttons are near edges; timeouts log out quickly.
- Form asks for CNIC, phone, DOB, address, and symptoms on one screen.
- Error messages use medical terms and acronyms.

- There is no 'Back' or 'Undo' during form entry.

PART D — User-Group Specific Evaluation (20 marks)

For EACH user group: propose an evaluation plan for a mobile app feature. Mention methods, tasks, data, metrics, adaptations.

Q12. (5 marks) Design an evaluation plan for **Seniors (older adults)**.

Q13. (5 marks) Design an evaluation plan for **People with disabilities (choose at least two: visual, motor, hearing, cognitive)**.

Q14. (5 marks) Design an evaluation plan for **Illiterate / low-literacy users**.

Q15. (5 marks) Design an evaluation plan for **Teenagers**.

PART E — Design from Scratch (Capstone) (10 marks)

Q16. (10 marks) Design the solution end-to-end based on the requirements below.

- You are designing a 'Campus Shuttle Tracker' app for a university. Users are students, seniors (visiting parents), and staff.
- Core screens required: (1) Onboarding + language selection, (2) Home map showing shuttle live location and ETA, (3) Route list and stop details, (4) 'Report an Issue' (crowding, delay, safety), (5) Notifications settings.
- Constraints: low connectivity, high sun glare outdoors, and first-time users unfamiliar with maps.
- Include: information architecture, interaction flow, and 8+ design decisions justified using named principles/heuristics.

MODEL ANSWERS + MARKING SCHEME

Self-check section. Marking scheme shows expected keywords + depth.

PART A — Model Answers (30 marks)

Q1: Usability testing + 3 goals (6 marks)

Usability testing is an evaluation method where we observe representative end users performing realistic tasks on a system/prototype and collect empirical evidence to improve usability. We capture both quantitative measures (task success, time, errors) and qualitative evidence (think-aloud, observed confusion, interview feedback) to guide redesign.

The three goals commonly emphasized are: (1) informing design (how users actually behave, what they expect, where they hesitate), (2) eliminating design problems and user frustration (detect breakdowns early and fix them before release), and (3) improving business or outcome metrics (higher completion, fewer support calls, improved retention/trust). These map to the usability dimensions effectiveness, efficiency, and satisfaction.

Marking Scheme

Key points	Marks
Correct definition: representative users + realistic tasks + observation/empirical data	2
Mention quant + qual metrics	1
3 goals listed correctly	3

Q2: Informative feedback vs visibility of system status (6 marks)

Both principles concern feedback, but their framing differs. Nielsen's 'visibility of system status' states the system should always keep users informed about what is going on, through timely, understandable feedback about current state: loading, processing, applied filters, network, mode, and progress.

Shneiderman's 'offer informative feedback' emphasizes that every user action should produce a meaningful response at an appropriate level of detail. It is action-response oriented: after a tap/click, show that the action worked, what changed, and what to do next, so the dialog makes sense.

Example (visibility): while searching for a driver, show live status 'Searching...', animation, and ETA updates. Example (informative feedback): after applying a voucher, show 'Voucher applied (Rs100 off)' and update the total with an option to remove it (supports reversal and control).

Marking Scheme

Key points	Marks
Explain visibility of system status correctly	2
Explain informative feedback correctly	2
Examples mapped to the correct rule names	2

Q3: Support internal locus of control (6 marks)

Supporting an internal locus of control means users feel they are initiating actions and the system responds predictably, rather than the system forcing them through unexpected states. This reduces anxiety and increases willingness to explore, which improves learnability and satisfaction.

Feature 1: clear 'Undo/Cancel/Back' and safe exits throughout flows (e.g., cancel a ride, undo removing an item). Combine with confirmation for destructive actions to preserve user control and freedom and make actions reversible.

Feature 2: provide review + edit before final submission (transfer, booking, checkout). Users can adjust details without restarting, and the system provides closure through a confirmation/receipt screen. This makes the dialog feel user-driven and predictable.

Marking Scheme

Key points	Marks
Definition of internal locus of control (user feels in control; predictable)	2
Two concrete features aligned to the principle	3
Impact explained (reduced anxiety, fewer catastrophic errors)	1

Q4: Reduce short-term memory load (6 marks)

This principle acknowledges the limits of working (short-term) memory. Interfaces should not require users to remember information across screens or steps. Instead, prefer recognition rather than recall: users should be able to recognize options and context on screen rather than recall them from memory.

Technique 1: make choices and history visible: recent orders, saved addresses, auto-complete, pre-filled forms, contextual hints/examples. Technique 2: keep key context visible during multi-step tasks using summaries, steppers, and inline validation so users don't have to remember what they entered.

Result: lower cognitive load, fewer slips and mistakes, faster completion—especially for novices, seniors, and low-literacy users.

Marking Scheme

Key points	Marks
Working memory limit + recognition vs recall explained	2
Two concrete techniques given	3
Impact explained	1

Q5: Iterative design + triangulation (6 marks)

Iterative design is a cycle: design → prototype → evaluate → redesign. We test early and often because usability problems are cheaper to fix before implementation, and each iteration improves effectiveness, efficiency, and satisfaction.

Triangulation means we do not treat one method or one small study as 'truth'. Different methods reveal different insights: expert reviews catch heuristic violations, usability tests reveal real breakdowns, and field studies reveal context (glare, interruptions, low connectivity). By combining evidence, we make stronger decisions.

In practice, triangulation prevents overfitting to one sample: we prioritize issues that repeat across methods and iterate until major problems are resolved.

Marking Scheme

Key points	Marks
Iterative design defined (cycle + repeated evaluation)	2
Triangulation defined (multiple methods/data)	2
Effect on decisions (stronger evidence, avoids overfitting)	2

PART B — Model Answers (20 marks)

Q6: Food Delivery Checkout Screen (10 marks)

- 1) Hidden 'Apply promo' under 'More options' → Recognition rather than recall + Visibility. Users cannot see the action and may forget it exists; lowers conversion and increases frustration.
- 2) 'Place Order' only at bottom → Efficiency of use + User control. Excess scrolling increases time-on-task, especially when users just want to confirm.
- 3) Too many equal-weight sections → Aesthetic and minimalist design (clutter). Important elements (total, payment, address) are not dominant; increases cognitive load.

- 4) Fee appears late (after pressing Place Order) → Match with real world + Informative feedback. Violates user expectation of transparent pricing; harms trust.
- 5) Back clears cart without warning → User control and freedom + Easy reversal. Catastrophic loss; users feel punished.
- 6) Error 'Code 14' → Help users recognize, diagnose, recover from errors. No explanation or next steps; user cannot recover.
- 7) No progress/step structure → Dialogs to yield closure. User doesn't know if they're almost done.
- 8) Optional extras mixed with essentials → Mental model mismatch; lack of hierarchy.
- 9) Total changes without breakdown near total → Informative feedback violated.
- 10) No inline validation until end → Error prevention violated.

Fixes (principle-driven): restructure into clear steps or strong hierarchy; persistent bottom bar with total + primary CTA; transparent fee breakdown before confirmation; make promo visible as a button; add undo/exit confirmation; improve error messages with causes and recovery actions; inline validation and saving state.

Marking Scheme

Key points	Marks
≥6 issues identified	3
Correct mapping to named principles/heuristics	4
Impact described	2
Concrete fixes tied to principles	1

Q7: University Portal Login + OTP (10 marks)

- 1) Ambiguous 'User ID' → Match with real world + Help. Add hint 'ERP / Email' and examples.
- 2) No password rules → Error prevention. Show requirements and inline validation.
- 3) No resend code → User control and freedom + Efficiency. Add resend after timer and 'change method'.
- 4) Expiry forces restart → Easy reversal violated. Allow staying in context and re-sending.
- 5) Remember me on OTP → Consistency and standards violated; move it to login.

- 6) Enter submits early and clears OTP → Error prevention violated; keep fields and submit only when complete.
- 7) Error far from focus → Visibility violated; show inline error and focus on field.
- 8) Unclear error messages ('invalid format') → Help users recover.
- 9) No loading state → Visibility; users may double submit.
- 10) Overall reduces locus of control → user feels trapped.

Fixes: clear labeling, examples, inline rules; resend + keep context; better keyboard handling; inline errors near fields; preserve input; add progress/loading feedback and confirmation/closure.

Marking Scheme

Key points	Marks
≥6 issues + correct mappings	6
Covers error prevention + recovery + control	2
Concrete fixes	2

PART C — Model Answers (20 marks)

Q8: Ride-hailing 'Schedule Ride' (5 marks)

This design is error-prone because the default time is unrealistic and AM/PM is a tiny control. It violates error prevention and recognition rather than recall, because users must carefully remember and verify AM/PM and the date.

Fix: use smart defaults (next available time), allow 24-hour format, and show the selected date and time in a large, readable summary row (visibility of system status). Make calendar selection explicit (not hidden behind a tiny icon).

Add a review/confirm screen with editable fields before scheduling, plus an immediate confirmation toast + receipt screen (closure). Surface 'cancel scheduled ride' in the scheduled ride card, not buried in settings (user control and freedom + easy reversal).

Keywords to include for full marks

- error prevention; recognition rather than recall; visibility; closure; user control and freedom; reversible actions; smart defaults; review screen

Q9: E-commerce returns form (5 marks)

The long dropdown increases cognitive load and reduces findability (aesthetic/minimalist design) and forces users to recall details. The requirement conflict (photo optional but demanded) is a classic consistency/error-prevention problem.

Fix: categorize reasons with short labels + examples; add search within reasons. Replace manual order number entry with a selectable list of recent orders (recognition rather than recall). Make photo truly optional or conditionally required based on reason, and label it clearly with 'why' (error prevention).

After submission, provide closure: return ID, instructions, and tracking. Provide informative feedback with status states ('Request received', 'Pickup scheduled', etc.) and clear error recovery guidance if submission fails.

Keywords to include for full marks

- reduce short-term memory load; recognition vs recall; error prevention; consistency; closure; informative feedback; conditional fields

Q10: Banking money transfer (5 marks)

Random beneficiary order and search-only-by-account number forces recall and increases error risk. Hiding fee/limit until late violates match with real world and transparency expectations in finance apps.

Fix: sort by recency/favorites and allow search by name + last digits (recognition). Show fees, limits, and total upfront (visibility, match with real world). Add a review step with 'Edit' before final submit (control + error prevention).

On failure, keep the user in context and preserve input; provide a human-readable error with retry and alternative actions. On success, show a receipt with reference number and share/download option (closure).

Keywords to include for full marks

- transparency; visibility; recognition; user control; error prevention; error recovery; closure; receipt/reference number

Q11: Hospital kiosk (5 marks)

The kiosk violates universal usability: small text, edge buttons, quick timeouts, high information density, and technical error messages. It creates high cognitive load and increases motor and visual errors.

Fix: large typography, high contrast, large spaced touch targets, and safe positioning of primary buttons. Break the form into steps with a progress indicator (reduce memory load, closure). Use plain language and examples; provide inline validation and help.

Allow 'Back' and 'Undo' on each step (user control and freedom). Use timeout warnings rather than sudden logouts; confirm completion with printed/SMS details (closure).

Keywords to include for full marks

- universal usability; accessibility; chunking; progress indicator; plain language; undo/back; timeout warning; error prevention

PART D — Model Answers (20 marks)

Q12: Seniors (older adults)

Focus: Readability, learnability, confidence, error recovery, trust; avoid time pressure.

Recommended methods: Moderated usability testing with think-aloud + short post-task interview; optionally pre-test questionnaire on device familiarity; consider field test under glare/noise.

Core tasks: Book feature end-to-end; change a detail; recover from a mistake (undo/cancel); find help; adjust accessibility settings.

Metrics: Success rate; time; errors; number of assists; backtracks; confidence rating; SUS; notes on confusion points.

Adaptations: Shorter sessions with breaks; larger font enabled; calm moderation; printed task cards; ensure participants use their typical device settings; reassure 'interface is being tested'.

Risks/limitations: Small samples; learning effects; health variability; lab environment differs from real context.

Output: Issue list with severity + redesign recommendations mapped to heuristics; prioritize barriers to completion and trust.

Q13: Disabilities (visual + motor example)

Focus: Accessibility barriers (labels, focus, contrast) and interaction barriers (touch target size, gesture alternatives, timeouts).

Recommended methods: Triangulation: expert accessibility inspection + heuristic evaluation + usability tests with assistive tech users; add constrained-input simulation if recruitment is hard.

Core tasks: Navigate to feature; complete main transaction; handle permissions; recover from errors; change settings; confirm final action.

Metrics: Completion, time, errors; screen-reader focus order issues; contrast failures; mis-taps; gesture failure rate; recovery success.

Adaptations: Use participants' own assistive tech; longer timeouts; allow breaks; ensure screen reader compatible; provide non-color cues; one-handed mode.

Risks/limitations: Recruitment difficulty; device diversity; simulated impairment isn't identical to lived experience.

Output: Accessibility audit log + prioritized fixes; ensure compliance and usability improvements with evidence.

Q14: Illiterate / low-literacy

Focus: Comprehension of icons/flow, reliance on recognition, fear of making mistakes, language and audio support.

Recommended methods: Contextual inquiry + moderated usability test; icon comprehension test; participatory design (choose icons/labels).

Core tasks: Complete task using landmarks; confirm action; understand feedback; recover from mistakes.

Metrics: Task success, misinterpretations, helps requested, retries, time; qualitative notes of confusion; confidence rating via pictorial scale.

Adaptations: Oral instructions; audio prompts; avoid written surveys; use emoji/picture rating; local language; allow companion support if realistic.

Risks/limitations: Hard to isolate literacy from overall digital familiarity; social desirability bias.

Output: Simplified icon set + redesigned navigation; recommendation for multimodal feedback and clearer hierarchy.

Q15: Teenagers

Focus: Speed, friction, engagement, personalization, privacy expectations; feature discovery.

Recommended methods: Rapid usability tests + A/B tests for micro-interactions; short survey/interview; log analysis if available.

Core tasks: Complete task quickly; customize; discover new feature; handle permissions; recover from mistake; share information safely.

Metrics: Completion/time/errors; rage taps; drop-off; discovery rate; satisfaction/trust score; preference between A/B variants.

Adaptations: Short tasks, allow exploration; test in realistic conditions; ensure privacy and ethical consent handling.

Risks/limitations: Peer influence in group sessions; novelty effect; device variability.

Output: Prioritized friction points + redesign for speed and clarity; improved privacy messaging; guidance for delight without sacrificing usability.

PART E — Model Answer (10 marks)

Q16: Campus Shuttle Tracker — model design (10 marks)

1) Information Architecture: Onboarding (language + accessibility presets) → Home (live map + ETA card + nearest stop) → Routes (list) → Stop details (arrivals, crowding, report) → Report issue (category + optional note) → Notifications (toggles). Keep essentials first, optional settings later (aesthetic/minimalist + mental model).

2) Flow: First launch chooses language. Home immediately shows nearest stop + next arrival with large readable text and clear status (visibility). Routes provides an alternative for users unfamiliar with maps (recognition). Stop details gives clear, structured info and a single primary action (set alert or report issue). Report ends with confirmation and ticket ID (closure).

3) Design decisions justified (write as bullet list in exam):

- High-contrast UI + large typography for outdoor glare → universal usability + accessibility.
- Persistent ETA/status card (not hidden behind tabs) → visibility of system status.
- Saved/favorite stops + recent stops → recognition rather than recall; reduces memory load.
- Route list with simple names matching signage (e.g., 'Main Gate → Hostel') → match with real world.
- Offline/low-connectivity mode: cache last known ETA and show 'last updated' timestamp → visibility + trust.

- Simple error states: 'No GPS', 'Network weak' with recovery steps → help users recover from errors.
- Report issue uses predefined categories + optional text (progressive disclosure) → error prevention + minimalist design.
- Confirm destructive actions (turning off alerts) and allow undo → user control and freedom.
- Consistent placement of primary CTA across screens → consistency and standards.
- Clear end-state after actions (ticket created, alert set) → dialogs to yield closure.

Marking Scheme

Key points	Marks
IA includes all required screens	3
Clear flow including constraints	2
≥8 justified decisions with correct principle names	4
Clarity about feedback/closure	1

APPENDIX A — User-Group Evaluation Question Bank (Frameworks)

Use these prompts to practice. For each, write a structured plan: Users → Method(s) → Tasks → Metrics → Adaptations → Risks → Output.

1. Seniors: Medication Reminder App

Evaluate 'set schedule' + 'mark as taken' and error recovery if they miss a dose.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

2. Visual impairment: Grocery App

Evaluate search + checkout using screen reader/zoom; focus on labels and focus order.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

3. Motor impairment: ATM kiosk

Evaluate withdraw cash with tremor; focus on touch target size, spacing, timeouts.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

4. Low literacy: Government subsidy form

Evaluate apply flow with icons/audio; focus on comprehension and confirmation.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

5. Teenagers: Study group app

Evaluate create group + invite + share notes; focus on speed, privacy permissions.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

6. Hearing impairment: Video learning app

Evaluate captions/transcripts and visual alerts; no audio-only cues.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

7. Cognitive impairment/ADHD: Task planner

Evaluate add task + reminder; focus on cognitive load and distractions.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

8. Mixed public kiosk: Railway ticket machine

Evaluate buy ticket for diverse population; universal usability under time pressure.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

9. First-time smartphone users: Wallet app

Evaluate top-up and pay; focus on learnability and trust.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

10. Color blindness: Analytics dashboard

Evaluate chart interpretation; avoid color-only encoding; add labels/patterns.

Model approach: Use moderated usability testing (or contextual inquiry) + tailored adaptations; include effectiveness/efficiency/satisfaction metrics; document issues with severity and link to heuristics.

APPENDIX B — Accessibility + Universal Usability Checklist

Visual

- Text size and scalable layout
- High contrast; avoid relying on color only
- Clear focus order/indicator
- Meaningful labels/alt text; readable errors

Motor

- Large touch targets + spacing
- Avoid precision-only gestures; provide alternatives

- Longer timeouts + warnings
- Undo for accidental taps; forgiving input

Hearing

- Captions/transcripts
- Visual alternatives to sound cues
- Vibration/visual alerts

Cognitive

- Reduce clutter; chunk steps
- Consistent layout/terminology
- Plain language; progressive disclosure
- Keep context visible (reduce memory load)

Environment

- Glare, motion, interruptions, low connectivity, noise

APPENDIX C — Heuristic-to-Fix Cheat Sheet (Exam keywords)

Visibility of system status

- Loading/progress; spinners
- State labels (selected date/time)
- Last updated timestamps
- Receipts/confirmations

User control and freedom

- Undo/cancel/back

- Clear exits without losing data
- Confirm destructive actions
- Editable review step

Recognition rather than recall

- Show recent items
- Auto-complete
- Visible options
- Inline examples/hints

Error prevention

- Input constraints/masks
- Inline validation
- Safe defaults
- Disable invalid actions

Help users recover from errors

- Plain language errors
- Explain cause
- Provide next steps
- Preserve user input

Consistency and standards

- Same terminology
- Standard button placement
- Consistent icons

- Platform conventions

Aesthetic and minimalist design

- Remove non-essential content
- Strong hierarchy
- Progressive disclosure
- Reduce visual noise

APPENDIX D — Extra Practice Exam (Second Paper)

Attempt in 90 minutes. No model answers included here so you can practice under pressure.

1. Explain direct manipulation and identify 'rapid, incremental, reversible actions' in a familiar app (mention the term).
2. Compare heuristic evaluation vs cognitive walkthrough (when to use each).
3. Given a dashboard with 12 charts, explain 6 issues under 'aesthetic and minimalist design' and propose fixes.
4. Design onboarding for a low-literacy user: language, icons, audio; justify with principles.
5. Create an evaluation plan for a public kiosk used by seniors and people with disabilities (triangulation required).
6. Fix a password reset flow that has unclear errors and forces restart; use named heuristics.

APPENDIX E — Glossary of High-Value HCI Keywords (Write these exact words)

Use these as 'definition sentences' in long answers. Keep them crisp and terminology-correct.

Direct manipulation: Interaction style where users act on visible objects with physical actions (drag, pinch, tap) and get immediate, continuous feedback; supports rapid, incremental, reversible operations.

Gulf of execution: Gap between the user's goals and the actions the interface makes available; widened when controls are hidden or unclear.

Gulf of evaluation: Gap between the system's state and the user's ability to perceive/interpret it; widened when feedback is missing or ambiguous.

Mental model: User's internal understanding of how a system works; good design matches and gradually shapes the mental model through consistent behavior and feedback.

Affordance: Perceived properties of an object that suggest possible actions; often strengthened by signifiers (labels, icons, handles).

Signifier: A perceivable cue that indicates where actions should take place (e.g., a button shape, a label, a handle).

Feedback: System response that informs the user that an action occurred and what the result is; includes visibility of system status and informative feedback.

Constraints: Design limits that prevent errors by restricting possible actions (input masks, disabled options, step order).

Error prevention: Designing to stop errors before they happen using constraints, safe defaults, and inline validation.

Recovery from errors: Helping users recognize, diagnose, and recover through clear messages, preserving input, and offering next steps.

Recognition vs recall: Prefer showing options/cues so users recognize what to do instead of remembering information from memory.

Cognitive load: Amount of mental effort required; reduced by chunking, hierarchy, consistency, and minimizing clutter.

Progressive disclosure: Reveal advanced/optional information only when needed, keeping initial screens simple.

Closure: Design dialogs to yield a clear end state (confirmation/receipt) so users feel completion.

Consistency and standards: Follow platform conventions and keep terms/layout stable so users can transfer learning.

Universal usability: Designing so diverse users (age, ability, literacy, devices) can use the system; includes accessibility.

Usability metrics: Effectiveness (success), efficiency (time/steps), satisfaction (perceived comfort/confidence).

Heuristic evaluation: Expert inspection method where evaluators check UI against heuristics and log issues with severity.

Cognitive walkthrough: Method focusing on learnability for first-time users; step through tasks asking if users know what to do and get feedback.

Think-aloud protocol: Users verbalize thoughts while completing tasks, revealing expectations and confusion.

Contextual inquiry: Field method: observe/interview users in their real context to understand workflow and constraints.

A/B testing: Compare two variants to see which performs better on defined metrics; good for micro-interactions at scale.

Low-fidelity prototype: Quick, cheap representation (paper, sketches) used early to explore structure and flow.

High-fidelity prototype: Detailed interactive prototype close to final UI used for realistic testing and stakeholder alignment.

Fitts's Law: Time to acquire a target depends on distance and size; implies bigger, closer buttons are faster to tap.

Hick's Law: Decision time increases with number/complexity of choices; implies simplify menus and group options.

Learnability: How easy it is for new users to accomplish basic tasks the first time they use the design.

Memorability: How easily users re-establish proficiency after not using the system for a while.

Efficiency: Speed and resource use once users have learned the design; often improved via shortcuts.

Satisfaction: How pleasant the experience feels; includes trust, comfort, and perceived control.

Mapping: Relationship between controls and their effects; good mapping reduces errors and supports mental models.

Visibility: Making relevant objects, actions, and system state perceptible; reduces gulfs and confusion.

Mode: A state where the same input yields different outcomes; risky if mode is hidden or unclear.

Slips vs mistakes: Slips are execution errors (intention correct); mistakes are goal/plan errors (mental model wrong).

Novice vs expert users: Novices need guidance and clarity; experts need efficiency, shortcuts, and flexibility.

Persona: Archetypal user profile representing goals, behaviors, and constraints; guides design decisions.

Scenario: Narrative of how a persona achieves a goal using the product; supports task-centered design.

Task analysis: Breaking down tasks into steps, decisions, and information needs to design better flows.

Accessibility: Designing so people with disabilities can perceive, understand, navigate, and interact; includes assistive tech.

WCAG: Guidelines for web accessibility: perceivable, operable, understandable, robust (POUR).

SUS: System Usability Scale: 10-item questionnaire producing a single usability score.

Severity rating: Prioritization of usability problems by impact and frequency (and persistence); used in heuristic eval.

Delight: Positive emotion from smooth interactions, micro-animations, and thoughtful defaults—without harming clarity.

Trust: User belief that system is safe and predictable; supported by transparency, consistent feedback, and reversible actions.

Learnability: How easy it is for new users to accomplish basic tasks the first time they use the design.

Memorability: How easily users re-establish proficiency after not using the system for a while.

Efficiency: Speed and resource use once users have learned the design; often improved via shortcuts.

Satisfaction: How pleasant the experience feels; includes trust, comfort, and perceived control.

Mapping: Relationship between controls and their effects; good mapping reduces errors and supports mental models.

Visibility: Making relevant objects, actions, and system state perceptible; reduces gulfs and confusion.

Mode: A state where the same input yields different outcomes; risky if mode is hidden or unclear.

Slips vs mistakes: Slips are execution errors (intention correct); mistakes are goal/plan errors (mental model wrong).

Novice vs expert users: Novices need guidance and clarity; experts need efficiency, shortcuts, and flexibility.

Persona: Archetypal user profile representing goals, behaviors, and constraints; guides design decisions.

Scenario: Narrative of how a persona achieves a goal using the product; supports task-centered design.

Task analysis: Breaking down tasks into steps, decisions, and information needs to design better flows.

Accessibility: Designing so people with disabilities can perceive, understand, navigate, and interact; includes assistive tech.

WCAG: Guidelines for web accessibility: perceivable, operable, understandable, robust (POUR).

SUS: System Usability Scale: 10-item questionnaire producing a single usability score.

Severity rating: Prioritization of usability problems by impact and frequency (and persistence); used in heuristic eval.

Delight: Positive emotion from smooth interactions, micro-animations, and thoughtful defaults—without harming clarity.

Trust: User belief that system is safe and predictable; supported by transparency, consistent feedback, and reversible actions.